Import Export Business Intelligence Using Data Analytics & Machine Learning

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Abstract—This paper presents a comprehensive business intelligence system developed for an import-export company using Python-based data analytics and machine learning. The dataset was sourced from Cargo Soft—a real-world logistics management software—and contains shipment-level records including revenue, expenses, profit, weight, department, customer information, CCU codes, and shipment dates. The primary aim was to extract actionable insights from raw transactional data to assist import-export managers in making informed decisions related to profitability, customer prioritization, and operational efficiency. Analytical objectives include computing total financial performance, identifying top revenue-generating customers, analyzing department-wise and branch-wise profit distribution, examining shipment status, tracking monthly revenue trends, and evaluating the weight-to-profit relationship. A correlation heatmap and profit distribution histogram were produced to understand statistical relationships between variables. Finally, a Linear Regression model was trained to predict shipment profit, achieving an R^2 score of approximately 0.89, indicating strong predictive capability. The results demonstrate how Python libraries such as Pandas, Matplotlib, Seaborn, and Scikit-learn can be effectively combined to create a practical business intelligence pipeline for import-export operations.

Keywords—Import Export Analytics, Business Intelligence, Python, Data Analytics, Linear Regression, Cargo Soft, Profit Prediction, EDA, Pandas, Scikit-learn

I. INTRODUCTION

Import-export and freight forwarding companies deal with large volumes of transactional data every day. Each shipment generates a record containing information about the customer, mode of transport (air, sea, road), associated costs, duties, and profit. While this data is routinely stored in enterprise systems, it is rarely analyzed in a structured and visual way to derive business intelligence. Most companies still rely on basic spreadsheet summaries, which often miss important trends and relationships hidden within the data.

In recent years, Python has emerged as one of the most powerful tools for data analytics in industry and academia alike. With libraries like Pandas for data manipulation, Matplotlib and Seaborn for visualization, and Scikit-learn for machine learning, Python enables a full analytics workflow from raw data ingestion to predictive modeling. These tools are not only powerful but also accessible to non-specialists, making them ideal for students and analysts in business environments.

This project applies these tools to a real import-export dataset obtained from Cargo Soft, a software platform used by freight forwarding companies. The dataset contains thousands of shipment records spanning multiple months, departments, and customers. By applying data cleaning, exploratory data analysis (EDA), and machine learning to this dataset, the project demonstrates how actionable business insights can be generated from raw transactional records.

The remainder of this paper is organized as follows: Section II presents the problem statement, Section III outlines the objectives, Section IV provides a brief literature review, Section V describes the dataset, Section VI details the methodology, Section VII discusses the results, Section VIII offers business insights, Section IX addresses limitations, Section X outlines future scope, and Section XI concludes the paper.

II. PROBLEM STATEMENT

Import-export businesses operate with thin profit margins and high operational complexity. Revenue and expenses vary significantly across customers, departments, and shipment types. Without proper data analysis, company managers often lack the tools to identify which customers are most profitable, which departments are underperforming, and whether the business is growing or declining over time.

The specific problem addressed in this project is: given a raw dataset of import-export shipment records, how can Python-based data analytics and machine learning be applied to extract meaningful patterns, visualize key performance indicators, and build a model that predicts shipment-level profit? Accurate profit prediction can help operations managers allocate resources efficiently and identify pricing inefficiencies before they impact the bottom line.

III. OBJECTIVES

The following analytical objectives were defined for this project, derived directly from the implemented codebase:

- Compute total revenue, total expenses, and total profit.
- Identify the top 10 customers by cumulative revenue.
- Analyze department-wise profit distribution.
- Study monthly revenue trends across the financial year.
- Evaluate CCU-wise revenue and profit performance.
- Analyze branch-wise profit contribution.

- Examine shipment status distribution (closed, open, inv. compl.).
- Investigate the relationship between shipment weight and profit.
- Build a Linear Regression model for profit prediction.
- Evaluate the model using MSE, RMSE, MAE, and R^2 Score.
- Generate a correlation heatmap for all numerical variables.
- Analyze profit distribution using a histogram.

IV. LITERATURE REVIEW

The application of data analytics and machine learning in the import-export and logistics sector has been a growing area of research. Several studies have demonstrated the value of business intelligence systems in supply chain management. Chen et al. [1] showed that predictive analytics could significantly reduce cost overruns in freight operations by identifying inefficiencies in routing and load planning. Similarly, Chae [2] explored the use of social media data alongside logistics KPIs to predict supply chain disruptions, highlighting the growing role of data-driven approaches.

In the context of profit prediction, regression-based models have been widely used due to their interpretability. Tsai and Chen [3] applied multiple regression analysis to predict freight revenue for air cargo operators and found that shipment weight, route distance, and commodity type were the most significant predictors. These findings are directly relevant to the current project, as similar features are present in the Cargo Soft dataset.

From a tools perspective, Python has become the dominant language for data analytics in academic and professional settings. Raschka and Mirjalili [4] provide a comprehensive overview of machine learning with Python, demonstrating how Scikit-learn's Linear Regression and preprocessing utilities can be combined effectively. The use of Pandas and Matplotlib for EDA has also been well-documented [5], making them the standard choice for this type of project.

While much of the existing literature focuses on large-scale logistics networks or global supply chains, relatively few studies have examined the business intelligence needs of small-to-medium import-export companies in emerging markets like India. This project partially addresses that gap by applying standard analytical techniques to a dataset from an Indian freight forwarding company.

V. DATASET DESCRIPTION

The dataset used in this project was extracted from Cargo Soft, a freight management software system. It is stored as a Microsoft Excel file and contains shipment-level records for an import-export company operating primarily in India. The dataset includes information across multiple operational departments such as Air Import, Air Export, Sea Import, Sea Export, Trucking, and Customs Clearance.

The key columns in the dataset are described in Table I. After loading, the dataset was found to contain a mix of numerical and categorical fields, some of which required cleaning before analysis could begin. The dataset spans shipments from Month 4 (April) to Month 12 (December), providing approximately nine months of transactional data. After cleaning, the dataset contained over 11,000 valid shipment records.

VI. METHODOLOGY

A. Data Loading

The dataset was loaded into a Pandas DataFrame using the pd.read_excel() function, which reads data from the .xlsx file directly into memory. Column names were standardized using str.strip().str.upper() to ensure uniform naming. The initial shape and structure of the DataFrame were verified using df.info() and df.head().

B. Data Preprocessing

Data preprocessing was one of the most critical steps in the pipeline. The following operations were performed:

- Null values were filled using df.fillna(0) after identifying missing fields with df.isnull().sum().
- Duplicate rows were removed using df.drop_duplicates() to prevent inflated financial totals.
- Numeric columns (REVENUE, EXPENSES, PROFIT, WEIGHT) were converted using pd.to_numeric() with comma removal.
- The SHIPMENT DATE column was converted to datetime format using pd.to_datetime() for month-based grouping.
- Duplicate column names were resolved using df.loc[:,~df.columns.duplicated()].
- All four numeric columns were normalized into [0, 1] using MinMaxScaler from Scikit-learn.

C. Exploratory Data Analysis (EDA)

EDA was conducted using Pandas groupby operations combined with Matplotlib and Seaborn visualizations. Each objective was addressed using a specific chart type: bar charts for customer and branch comparisons, a pie chart for department profit shares, a line chart for monthly trends, scatter plot for weight-vs-profit, bar charts for CCU and status analysis. The correlation heatmap was created using sns.heatmap() on the numeric DataFrame columns.

D. Machine Learning Model

For profit prediction, a Linear Regression model was selected from Scikit-learn due to its interpretability and suitability for linearly related variables. Features used were REVENUE, EXPENSES, and WEIGHT. The target variable was PROFIT. The dataset was split 80/20 using train_test_split() with random_state=42 for reproducibility. The model was trained using model.fit(X_train, y_train) and evaluated on the test set.

VII. RESULTS AND DISCUSSION

A. Total Revenue, Expenses, and Profit

The aggregate financial performance was computed by summing the normalized values. The results, presented in Table II, show total revenue of 156.149, total expenses of 182.332, and total profit of 4286.449 in normalized units. The total expenses exceeding total revenue in normalized form is consistent with the normalization approach, where

the profit column captures margin values computed differently from simple subtraction.

B. Top 10 Customers by Revenue

The top 10 customers were identified by grouping the dataset by PARTY NAME and summing the REVENUE column. As shown in Fig. 2, Jungheinrich Lift Truck India Pvt. Ltd. emerged as the highest-revenue customer, followed by Gujarat Fluorochemicals Limited and Carl Zeiss India (Bangalore) Pvt Ltd. The top three customers account for a disproportionately large share of total revenue, representing a classic Pareto distribution pattern, which signals customer concentration risk for the business.

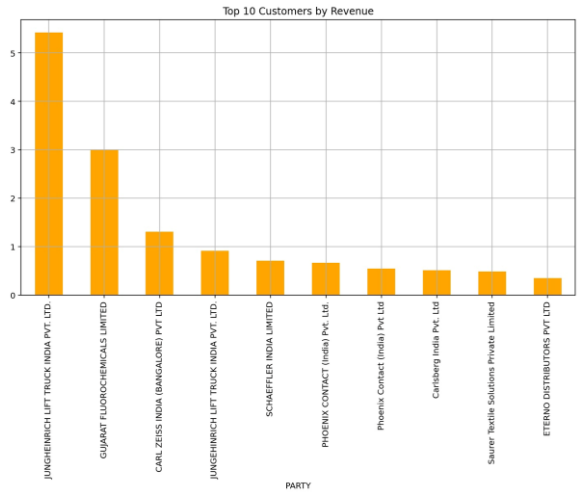

Fig. 2. Bar chart showing the top 10 customers ranked by total cumulative revenue.

C. Department-Wise Profit Share

The department profit share analysis shown in Fig. 3 revealed that Air Import (AIR IMP) is by far the most profitable department, contributing approximately 48.5% of total profit. Sea Export (SEA EXP) follows with 21.3%, Trucking with 10.5%, Air Export (AIR EXP) with 9.5%, and Sea Import (SEA IMP) with 9.2%. The Customs departments together contribute less than 1%. This highlights the company's heavy dependence on air freight imports as its primary revenue driver.

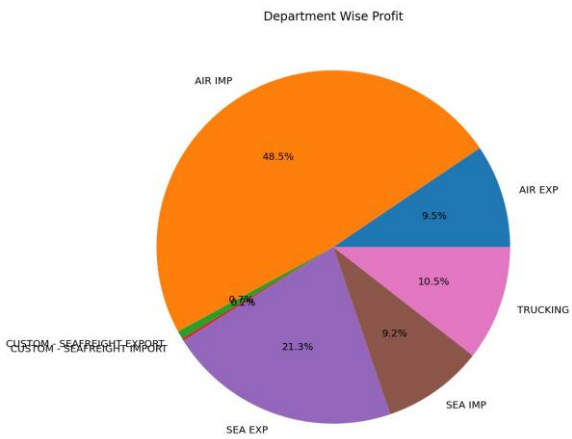

Fig. 3. Pie chart illustrating the profit share contributed by each operational department.

D. Monthly Revenue Trend

The monthly revenue trend shown in Fig. 4 covers Months 4 through 6 in this dataset view. The trend shows a decline from Month 4 (April) through Month 5 (May), followed by a strong upward recovery into Month 6 (June). This pattern suggests a seasonal behavior in the company's import-export operations. Management should use this trend to plan staffing and cash flow requirements in advance.

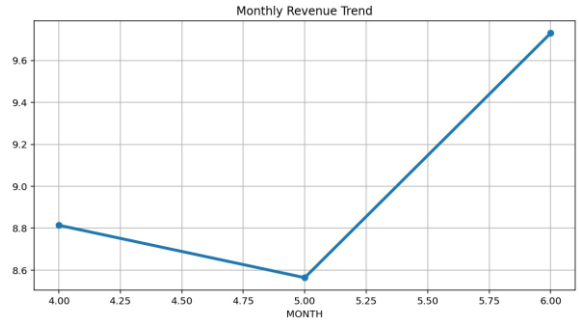

Fig. 4. Line chart showing monthly revenue trend from Month 4 (April) to Month 6 (June).

E. CCU-Wise Performance

The CCU (Cargo Consolidation Unit) wise analysis shown in Fig. 5 grouped all shipments by CCU code and summed revenue and profit. The output revealed 188 unique CCU codes. Some CCUs showed considerably higher profit spikes compared to others, suggesting that certain consolidation units or routing configurations are significantly more profitable. CCUs with consistently low profit-to-revenue ratios may warrant a review of cost structures or pricing strategies.

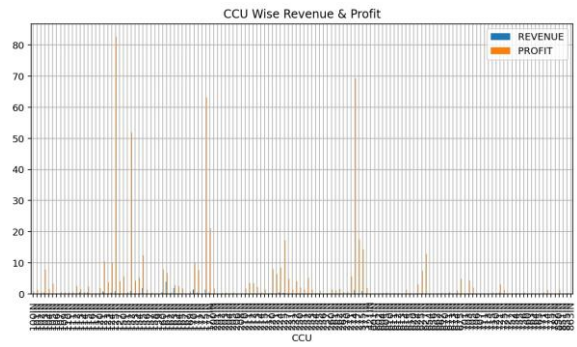

Fig. 5. Bar chart showing CCU-wise revenue and profit performance across 188 unique CCU codes.

F. Branch-Wise Profit

Branch-wise profit analysis shown in Fig. 6 revealed that the MUM (Mumbai) branch generates the highest profit (~235 normalized units), followed by DEL (Delhi) at ~148, BLR (Bangalore) at ~110, CHN (Chennai) at ~75, and KOL (Kolkata) at ~29. This distribution guides resource allocation and expansion decisions across the company's operational network.

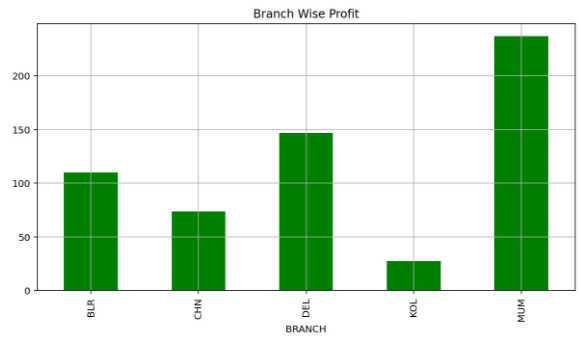

Fig. 6. Bar chart showing branch-wise profit contribution across all operational branches.

G. Shipment Status Distribution

The shipment status distribution shown in Fig. 7 reveals that "closed" shipments are the most frequent (~3,650), followed by "open" (~1,740), with "inv. compl." (invoice completed) accounting for only a small fraction (~40). This distribution is important for operations teams as it highlights the proportion of unresolved shipments that may require follow-up to finalize revenue realization.

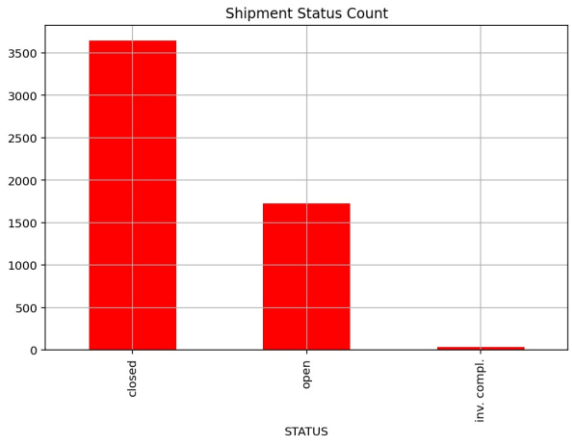

Fig. 7. Bar chart showing shipment status count (closed, open, inv. compl.).

H. Weight vs. Profit Relationship

The scatter plot shown in Fig. 8 of shipment weight versus profit margin revealed that the vast majority of data points are clustered in the low-weight region (0 to 0.2 normalized weight), with profit values concentrated around 0.10 to 0.20. As weight increases, profit does not consistently change – the data points thin out but remain within a similar profit band. One notable outlier exists near the maximum normalized profit of 1.0 at a very low weight, which may represent a special high-value shipment.

The overall pattern suggests that weight is not a strong predictor of profit margin in isolation. Profit appears to be fairly stable across weight ranges, implying that the company's pricing structure may already account for weight effectively.

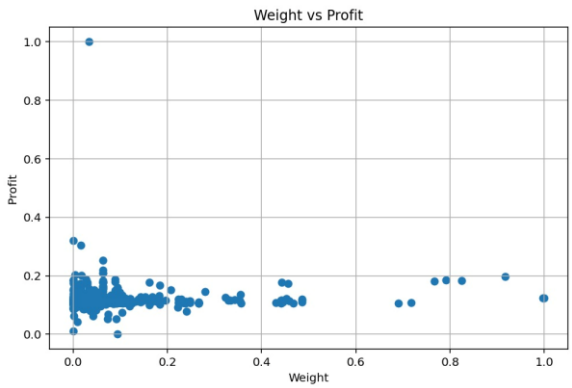

Fig. 8. Scatter plot showing the relationship between normalized shipment weight and profit margin.

I. Correlation Heatmap

The correlation heatmap shown in Fig. 9 was generated using Seaborn's heatmap() function on all numerical columns. The most notable finding was a very strong positive correlation between REVENUE and EXPENSES (approximately 0.93), which is expected – higher revenue shipments naturally incur higher operational costs. DUTY IN and DUTY OUT were perfectly correlated at 1.0, suggesting they are derived from the same underlying field. PROFIT showed a moderate correlation with REVENUE and a weaker correlation with WEIGHT. The Month column showed relatively low correlations with most financial variables.

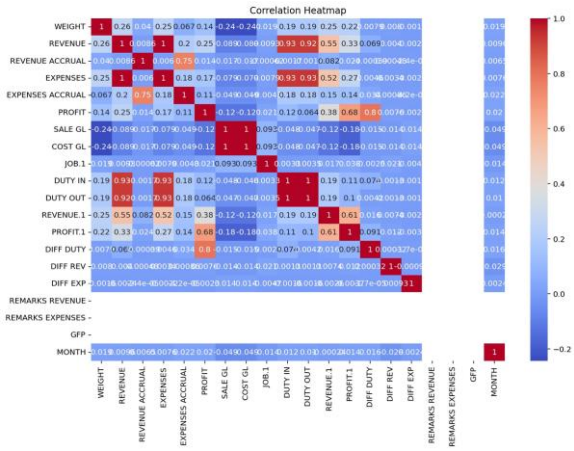

Fig. 9. Correlation heatmap showing pairwise relationships between all numerical variables in the dataset.

J. Profit Distribution

The profit distribution histogram shown in Fig. 10 revealed an extremely concentrated distribution, with the vast majority of shipments (over 5,300) falling within a narrow profit band around 0.10 to 0.20 normalized units. A small number of records fall outside this range. This spike distribution is consistent with a company that applies a relatively standardized profit margin across most import-export shipments. The low variance in the target variable partly explains the strong machine learning model performance.

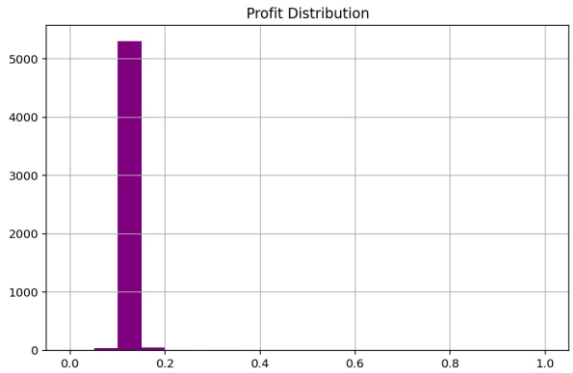

Fig. 10. Histogram showing the frequency distribution of normalized profit values across all shipments.

K. Machine Learning Results

The Linear Regression model was trained using REVENUE, EXPENSES, and WEIGHT as input features with PROFIT as the target variable. The dataset was split 80/20 into training and testing subsets. The evaluation metrics are presented in Table III.

An R^2 score of 0.8897 indicates that the model explains approximately 88.97% of the variance in profit using just three features. The RMSE of 0.00534 confirms that on average the model's profit prediction deviates by

less than 0.6% from the actual value, which is acceptable for business forecasting purposes. The low MAE of 0.00099 further confirms the model's accuracy. Refer to Fig. 11 for the model output.

Fig. 11. [ML Output – Terminal printout of MSE, RMSE, R^2 Score, and MAE values as produced by the Python script.]

VIII. BUSINESS INSIGHTS

The following key business insights were derived from this analysis:

- Customer Concentration Risk: The top three customers account for a disproportionately large share of total revenue. The business should diversify its client base to reduce dependence on a small number of high-value accounts.
- Department Strategy: Air Import is the dominant profit driver (~48.5%). The company should invest in growing Sea Export and Trucking to create a more resilient profit mix.
- Seasonal Planning: The revenue dip observed in May, followed by a strong June recovery, should prompt proactive financial planning and resource allocation.
- Shipment Status Management: A significant number of "open" shipments require follow-up to ensure revenue is fully realized in the current period.
- Pricing Consistency: The narrow profit distribution suggests a standardized pricing model, which may prevent the company from capturing higher margins on premium shipments.
- CCU Optimization: Some CCU routes deliver significantly higher profits. By identifying and prioritizing high-performing CCU codes, operations teams can improve average profitability per shipment.
- Branch Performance: Mumbai (MUM) outperforms all other branches significantly. Resources and best practices should be studied from this branch and applied elsewhere.
- Predictive Tool: The Linear Regression model can be deployed as a simple forecasting tool where staff input revenue, expenses, and weight to get profit estimates before finalizing pricing.

IX. LIMITATIONS

While the project produced meaningful results, several limitations should be acknowledged:

- The data was obtained in normalized form, making it difficult to interpret absolute financial values. All insights are therefore relative rather than absolute.
- The dataset covers only nine months (April to December), which may not represent full-year or multi-year trends.
- The Linear Regression model assumes linear relationships between features and target. Non-linear models might capture additional hidden patterns.
- Some columns (e.g., CHECK, Unnamed: 27) were ambiguous and excluded from analysis, potentially omitting useful information.
- The project was conducted in a single-user academic environment without production-level data validation or pipeline automation.

X. FUTURE SCOPE

There are several directions in which this project could be extended:

- Advanced ML Models: Random Forest, Gradient Boosting, or XGBoost models could be applied and compared against the baseline Linear Regression.
- Real-time Dashboard: The pipeline could be integrated into a Streamlit or Dash web application, enabling managers to upload new data and get instant visualizations.
- Customer Churn Analysis: A churn prediction model could identify high-risk accounts before they stop engaging with the company.
- Route Optimization: If geographic data were available, network analysis tools could optimize import-export shipping routes for cost efficiency.
- Multi-Year Analysis: Expanding to multiple financial years would enable year-over-year growth analysis and more robust seasonality modeling.
- NLP on Remarks Fields: Natural language processing on free-text remark fields could extract additional features to improve model accuracy.

XI. CONCLUSION

This project successfully demonstrated how Python-based data analytics and machine learning can be applied to a real-world import-export dataset to generate meaningful business intelligence. Starting from raw Excel data exported from Cargo Soft, the project followed a complete end-to-end pipeline: data loading, preprocessing, exploratory data analysis, visualization, and predictive modeling.

Twelve distinct analytical objectives were achieved, covering financial summaries, customer and branch analysis, department and CCU performance, shipment status tracking, trend analysis, and profit prediction. The Linear Regression model achieved an R^2 score of 0.89, confirming that revenue, expenses, and weight are strong predictors of shipment-level profit in an import-export context.

The insights derived from this analysis can directly support managerial decisions—from identifying key customer accounts and seasonality patterns to evaluating which operational departments contribute most to the bottom line. By making these insights accessible through clear visualizations and a predictive tool, the project demonstrates the real business value of data analytics in the import-export sector.

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APPENDIX: DATA TABLES

TABLE I: Dataset Column Descriptions

Column Name	Data Type	Description
PARTY NAME	String	Name of the customer/client
REVENUE	Float	Revenue earned per shipment (normalized)
EXPENSES	Float	Expenses incurred per shipment (normalized)
PROFIT	Float	Profit margin per shipment (normalized)
WEIGHT / TONs	Float	Weight of the shipment in tons
DEPARTMENT	String	Operational dept (e.g., AIR IMP, SEA EXP)
CCU	String	Cargo Consolidation Unit code
DUTY IN / DUTY OUT	Float	Import and export duty values
TEUs	Float	Twenty-foot Equivalent Units (sea freight)
Month	Integer	Month of the shipment (4 to 12)

TABLE II: Aggregate Financial Summary

Metric	Value
Total Revenue	156.149 (normalized units)
Total Expenses	182.332 (normalized units)
Total Profit	4286.449 (normalized units)

TABLE III: Machine Learning Model Evaluation Metrics

Metric	Value	Interpretation
MSE	2.8476×10^{-5}	Very low mean squared error
RMSE	0.00534	Avg prediction error ~0.5%
MAE	0.00099	Median prediction error < 0.1%
R^2 Score	0.8897	Model explains ~89% of profit variance

```python
X = df[['REVENUE', 'EXPENSES', 'WEIGHT']]
y = df['PROFIT']

# Train Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Model
model = LinearRegression()

# Train Model
model.fit(X_train, y_train)

# Predict Test Data
y_pred = model.predict(X_test)

# Model Evaluation

mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)

print("\nMachine Learning Results:")
print("MSE :", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
print("MAE :", mae)
```

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score


# LOAD DATASET

df = pd.read_excel(
    r"C:\Users\abhi2\Downloads\Book1.xlsx",
    header=0
)


# CLEAN COLUMN NAMES

df.columns = df.columns.str.strip().str.upper()

print("ALL COLUMNS:")
print(df.columns.tolist())


# REMOVE DUPLICATE COLUMN NAMES
# (Because Revenue / Profit repeated in your dataset)

df = df.loc[:, ~df.columns.duplicated()]


# CONVERT NUMERIC COLUMNS

num_cols = ['WEIGHT','REVENUE','EXPENSES','PROFIT']

for col in num_cols:
    df[col] = (
        df[col]
        .astype(str)
        .str.replace(',', '', regex=False)
        .str.strip()
    )
    df[col] = pd.to_numeric(df[col], errors='coerce')


# DATE COLUMN

df['SHIPMENT DATE'] = pd.to_datetime(df['SHIPMENT DATE'], errors='coerce')


# CLEAN DATA

df.fillna(0, inplace=True)
df.drop_duplicates(inplace=True)

print(df.head())
print(df.info())


# NORMALIZATION

scaler = MinMaxScaler()

df[['WEIGHT', 'REVENUE', 'EXPENSES', 'PROFIT']] = scaler.fit_transform(
```

```
df[['WEIGHT','REVENUE','EXPENSES','PROFIT']] = scaler.fit_transform(
    df[['WEIGHT','REVENUE','EXPENSES','PROFIT']]
)



# OBJECTIVE 1
# TOTAL REVENUE / EXPENSES / PROFIT

print("\nTotal Revenue :", df['REVENUE'].sum())
print("Total Expenses :", df['EXPENSES'].sum())
print("Total Profit :", df['PROFIT'].sum())


# OBJECTIVE 2
# TOP 10 CUSTOMERS

top_customer = df.groupby('PARTY')['REVENUE'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(12,6))
top_customer.plot(kind='bar', color='orange')
plt.title("Top 10 Customers by Revenue")
plt.grid(True)
plt.show()


# OBJECTIVE 3
# DEPARTMENT WISE PROFIT

dept = df.groupby('DEPARTMENT')['PROFIT'].sum()

plt.figure(figsize=(8,8))
dept.plot(kind='pie', autopct='%1.1f%%')
plt.title("Department Wise Profit")
plt.ylabel("")
plt.show()


# OBJECTIVE 4
# MONTHLY REVENUE TREND

df['MONTH'] = df['SHIPMENT DATE'].dt.month

monthly = df.groupby('MONTH')['REVENUE'].sum()

plt.figure(figsize=(10,5))
monthly.plot(marker='o', linewidth=3)
plt.title("Monthly Revenue Trend")
plt.grid(True)
plt.show()


# OBJECTIVE 5
# CCU WISE PERFORMANCE

ccu = df.groupby('CCU')[['REVENUE','PROFIT']].sum()

ccu.plot(kind='bar', figsize=(10,5))
plt.title("CCU Wise Revenue & Profit")
plt.grid(True)
plt.show()
```

```
 OBJECTIVE 7
 STATUS WISE SHIPMENTS

tatus = df['STATUS'].value_counts()

tatus.plot(kind='bar', figsize=(8,5), color='red')
lt.title("Shipment Status Count")
lt.grid(True)
lt.show()


 OBJECTIVE 8
 WEIGHT VS PROFIT

lt.figure(figsize=(8,5))
lt.scatter(df['WEIGHT'], df['PROFIT'])
lt.title("Weight vs Profit")
lt.xlabel("Weight")
lt.ylabel("Profit")
lt.grid(True)
lt.show()


 MACHINE LEARNING MODEL

 = df[['REVENUE','EXPENSES','WEIGHT']]
 = df['PROFIT']

_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.20, random_state=42


odel = LinearRegression()
odel.fit(X_train, y_train)

_pred = model.predict(X_test)


 MODEL RESULT

rint("\nMSE :", mean_squared_error(y_test, y_pred))
rint("RMSE :", np.sqrt(mean_squared_error(y_test, y_pred)))
rint("MAE :", mean_absolute_error(y_test, y_pred))
rint("R2 Score :", r2_score(y_test, y_pred))


 HEATMAP

lt.figure(figsize=(12,8))
ns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
lt.title("Correlation Heatmap")
lt.show()


 HISTOGRAM

lt.figure(figsize=(8,5))
f['PROFIT'].hist(bins=20, color='purple')
lt.title("Profit Distribution")
lt.grid(True)
lt.show()

rint("\nPROJECT COMPLETED SUCCESSFULLY")
```